

NET731 – Cloud Networking

Week 10 Practical: Final Presentation

Student Name: Ian Strauss

Student Number: 20240049

Slide 1 — Title Slide

NET731 – Cloud Networking

Final Azure Project Presentation

- Ian Strauss
 - 20240049
 - Module: NET731
 - Date: 12 May 2026
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Slide 2 — Project Overview

Project Overview

Environment Built

- Azure cloud networking environment
- Virtual machine deployment
- Virtual networking configuration
- Security implementation
- Monitoring and cost management

Purpose

- Learn practical Azure cloud administration
- Configure secure cloud infrastructure
- Monitor and optimise cloud resources
- Gain experience using Azure services

Screenshot Space

Name	Type	Location	Resource Group	Subscription	Last accessed
20240049storage	Storage account	UAE North	CTU-20240049-...	Azure for Stude...	19 minutes ago
vm-week6	Virtual machine	UAE North	CTU-20240049-...	Azure for Stude...	20 minutes ago
tb-vnet-01	Virtual network	Italy North	TechBridge-RG	Azure for Stude...	20 minutes ago
NSG-20240049	Network securit...	Italy North	CTU-20240049-...	Azure for Stude...	22 minutes ago
TechBridge-NSG	Network securit...	Italy North	TechBridge-RG	Azure for Stude...	25 minutes ago
LogforVM	Log Analytics w...	Italy North	CTU-20240049-...	Azure for Stude...	8 days ago
CTU-20240049-RG	Resource group		CTU-20240049-...	Azure for Stude...	8 days ago
Azure for Students	Subscription			Azure for Stude...	11 days ago
TechBridge-RG	Resource group	South Africa N...	TechBridge-RG	Azure for Stude...	29 days ago

Slide 3 — Architecture Diagram

Azure Architecture Diagram

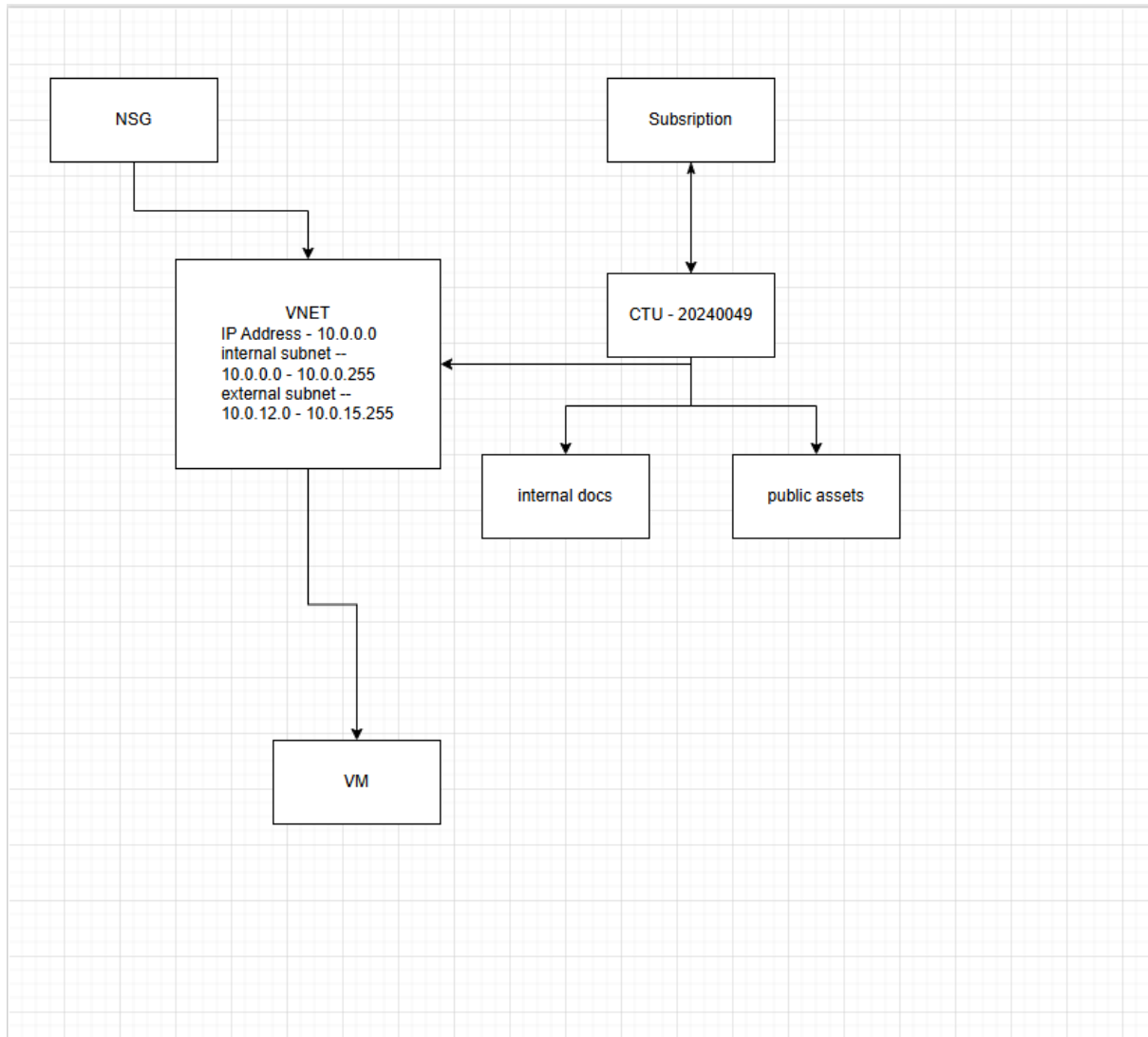
Components Included

- Resource Groups
- Virtual Network (VNet)
- Subnets
- Network Security Groups (NSGs)
- Virtual Machine
- Storage Account
- Azure Monitor
- Log Analytics Workspace

Explanation

The architecture diagram shows how networking, security, monitoring, and storage resources are connected within the Azure environment.

Screenshot Space



Slide 4 — Key Technical Decisions

Key Technical Decisions

Decision 1 — VM Size

- Selected B1s VM size
- Lower cost option
- Suitable for testing environment

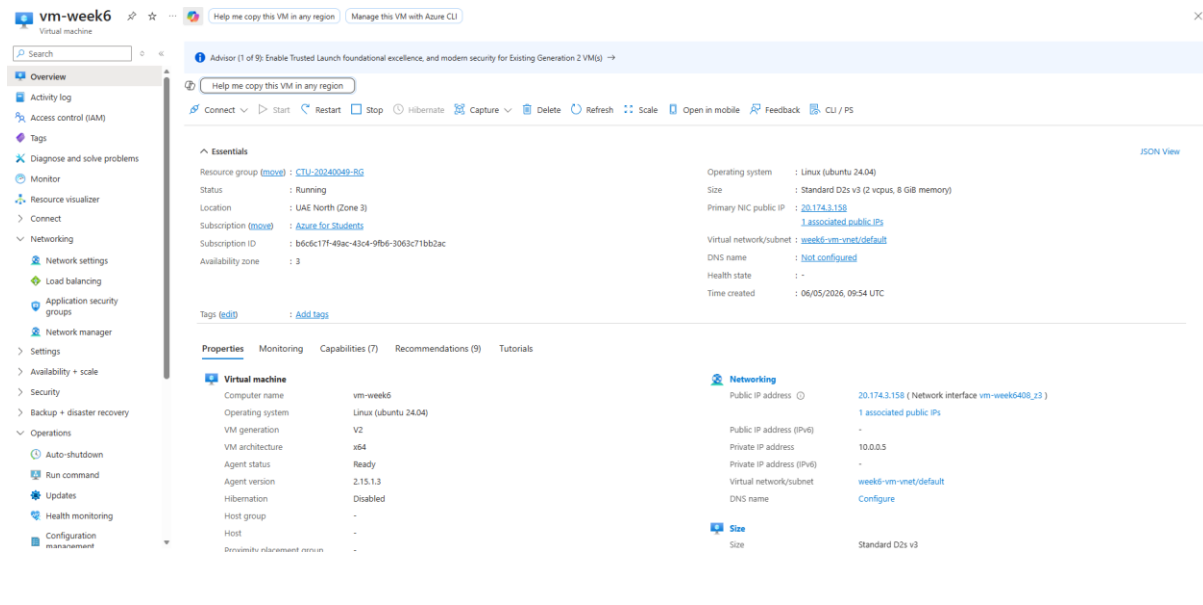
Decision 2 — NSG Configuration

- Configured inbound RDP rule
- Restricted access where possible
- Improved network security

Decision 3 — Monitoring Setup

- Enabled Azure Monitor
- Configured Log Analytics Workspace
- Allowed monitoring of performance metrics

Screenshot Space



Slide 5 — Security and Monitoring

Security and Monitoring

Security Implemented

- Network Security Groups (NSGs)
- RBAC permissions
- Defender for Cloud recommendations
- Access control configuration

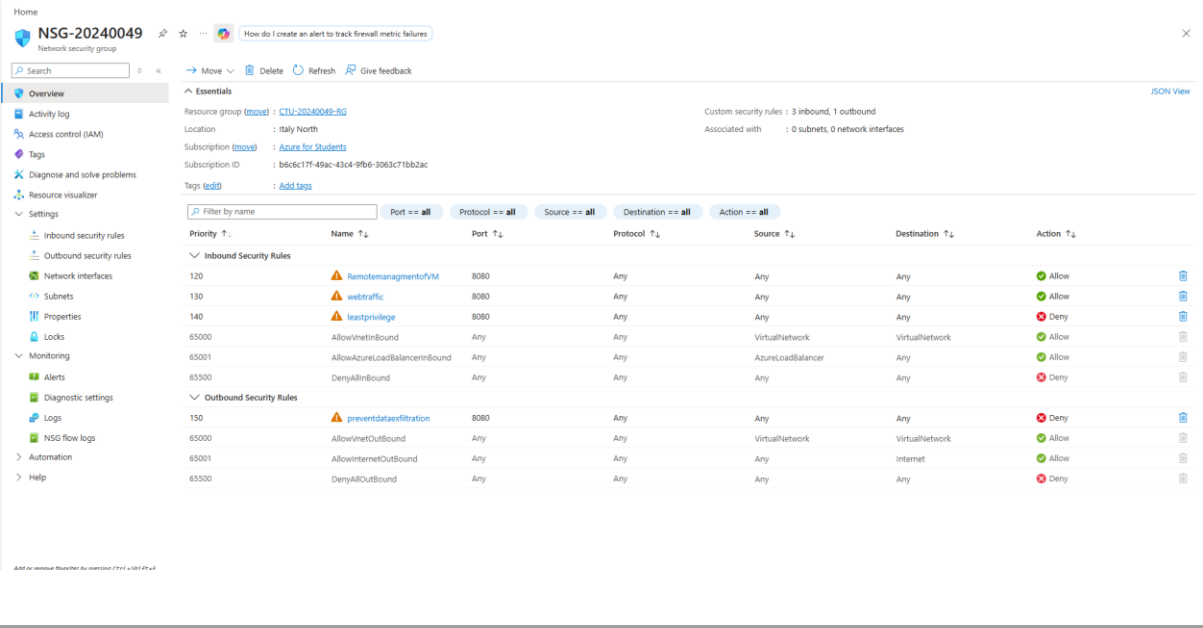
Monitoring Implemented

- Azure Monitor alerts
- CPU and network monitoring
- Budget alerts
- Log Analytics Workspace

Why It Was Important

These tools helped improve security, monitor system health, and detect performance or cost issues early.

Screenshot Space



Slide 6 — Cost Management

Cost Management

Cost Control Measures

- Created Azure budget alerts
- Monitored spending using Cost Analysis
- Stopped unused virtual machines
- Used low-cost VM sizes

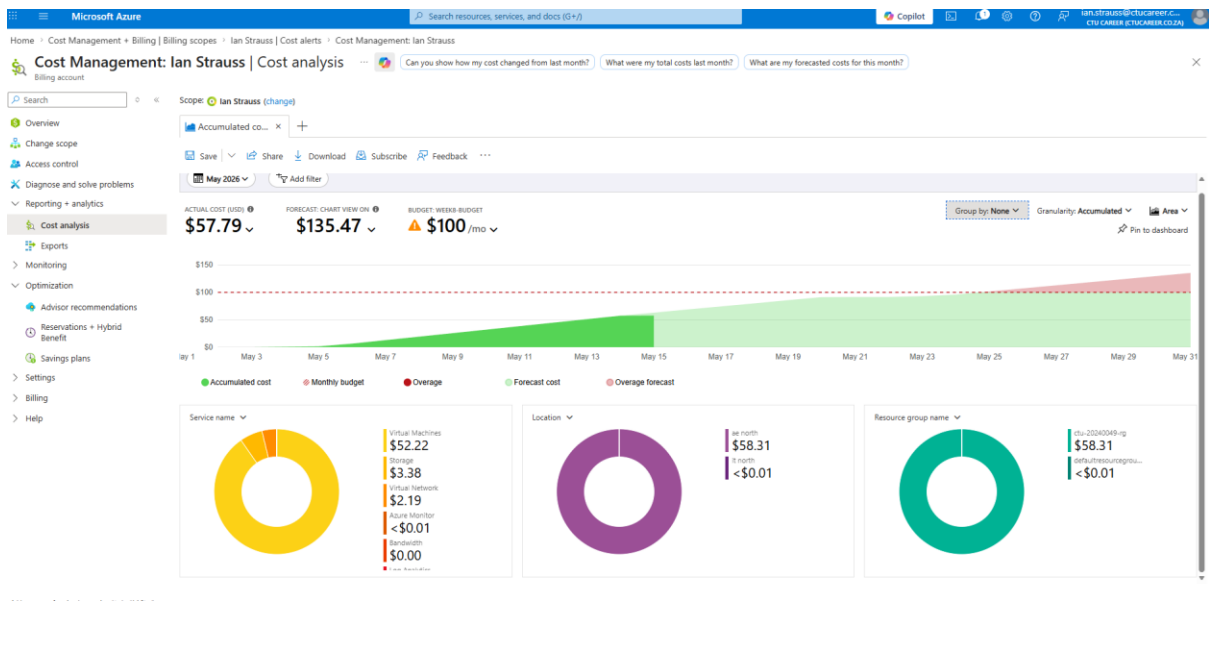
Projected Spend

- Estimated spend remained below \$100
- Virtual machine was the highest-cost resource

Lessons Learned

Cloud resources can become expensive quickly if not monitored properly.

Screenshot Space



Slide 7 — Challenges and Solutions

Challenges and Solutions

Challenge 1

Unable to deploy some VM sizes

Solution

Changed VM region and selected alternative VM sizes

Challenge 2

Connectivity and NSG issues

Solution

Reviewed NSG inbound rules and corrected RDP settings

Challenge 3

Managing Azure costs

Solution

Stopped and deallocated unused resources

Slide 8 — Lessons Learned

Lessons Learned

Skills Developed

- Azure networking
- VM deployment
- Cloud monitoring
- Security configuration
- Cost optimisation
- GitHub documentation

What I Would Do Differently

- Plan resource deployment earlier
- Organise screenshots weekly
- Use automation tools where possible

Slide 9 — Conclusion

Conclusion

Key Takeaways

- Successfully deployed an Azure cloud environment
- Configured networking, security, and monitoring
- Managed cloud spending effectively
- Improved practical cloud administration skills

Final Reflection

The project provided valuable real-world experience using Microsoft Azure and cloud networking technologies.

Thank You

Screenshots To Include

- ✓ Azure Dashboard
- ✓ Resource Group
- ✓ VM Overview

- ✓ NSG Rules
- ✓ Architecture Diagram
- ✓ Azure Monitor
- ✓ Cost Analysis
- ✓ Defender for Cloud